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Are schools with a better work environment better at retaining teachers? A repeated cross-sectional study examining schools' organisational and social risk factors and teacher retention

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ABSTRACT

This longitudinal study examines the relationship of psychosocial safety climate and psychosocial work factors with teacher turnover rates at 12- and 24-months follow-ups using objective turnover data from 14 schools in two Swedish municipalities. The study used an open cohort repeated cross-sectional research design and merged teachers' self-reported survey data on psychosocial safety climate and psychosocial work environment factors, which were aggregated to the school level, during the first school year ($n = 440$) and in the second school year ($n = 406$), to investigate their associations with annual school-level turnover data for each school year. Given the micro–macro data situation and the few higher-level groups ($n = 14$ schools), school-level data were analysed using disaggregation and hierarchical regression. The findings show that differences in teacher turnover between schools were mainly explained by municipal effects and labour market dynamics rather than differences in schools' psychosocial safety climate and psychosocial work factors.

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Psychosocial safety climate; Teacher turnover; Teacher retention; Psychosocial work factors; Micro–macro data

1. Introduction

In accordance with the Work Environment Act and its provisions, all schools in Sweden must systematically conduct organisational and social risk assessments to promote healthy work environments that safeguard employees' health and safety (Swedish Work Environment Authority, 2015). The objective of managing psychosocial risk factors in schools is to identify and intervene in psychosocial risk factors which can negatively impact the health and safety of school staff (Wischlitzki et al., 2020). Teachers are exposed to a range of psychosocial risk factors in their work environment such as high administrative burdens, student discipline issues, and time pressure, which take a toll on their work-related health if they are not addressed and managed (Arvidsson et al., 2019). The management of psychosocial risk factors in schools has received considerable attention due to the high levels of work-related ill health in the teaching profession, which have

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been attributed to the continuous exposure to untenable levels of psychosocial risk factors in the work environment (The Swedish Work Environment Authority, 2020, 2022). A report by the Swedish Work Environment Authority assessing the systematic work environment management in schools between 2013 and 2016 showed that nine out of ten schools had high levels of job demands and inadequate procedures for the identification, management, and prevention of psychosocial risk factors (The Swedish Work Environment Authority, 2017). The research literature is privy to these issues, which has provided the impetus for improving the implementation of systematic work environment management practices in schools (Kwak et al., 2019; Toropova et al., 2022, 2023). This is of particular importance for schools as challenging working conditions and work-related ill health are among the factors contributing to the vocational crisis in the teaching profession (European Commission, 2021).

Studies using occupational stress models such as the Job Demands-Resource model (JD-R) have shown that the interplay between teachers' exposure to psychosocial risk factors and their work-related ill health further extends to teachers' career decisions (Collie, 2023; Rajendran et al., 2020; Skaalvik & Skaalvik, 2020). The relationship between teachers' exposure to psychosocial risk factors and teacher turnover has been extensively studied due to the deleterious effects of teacher turnover on schools' organisational functioning and instructional quality (Gibbons et al., 2021; Sorensen & Ladd, 2020). Although teacher turnover is influenced by structural factors such as accountability and performativity pressures (Perryman & Calvert, 2019), labour market conditions (Falch et al., 2009; Falch & Strøm, 2005), and student composition factors (Karbownik, 2020), it is also strongly tied to the health-impairing influence of psychosocial risk factors in teachers' work environment (Madigan & Kim, 2021; Statistiska centralbyrån, 2017, 2023). The findings from the teacher turnover literature suggest that the mechanisms underlying teacher retention align with the objectives of psychosocial risk management and the safeguarding of teachers' work-related health (Geiger & Pivovarova, 2018; Madigan & Kim, 2021). However, few studies have examined the management of psychosocial risk factors in teachers' work environment and how it in turn relates to teacher retention (Wischlitzki et al., 2020). Most of the research in the teacher retention literature is conducted on the individual level and focuses on identifying protective job resources, such as collegial support, that can safeguard teachers from the health-impairing influence of psychosocial risk factors in the work environment and facilitate teacher retention (e.g. Pomaki et al., 2010). Although this provides an insight into the bottom-up process by which teachers can influence their work environment using proactive strategies, it disregards the systematic work environment management practices that the school leadership must adhere to. Longitudinal evidence has shown that school principals' leadership practices matter when it comes to teachers' perceptions of their work environment and their work-related health, which in turn affect their career decisions (Böhlmark et al., 2016; Burkhauser, 2016). However, a notable limitation of these studies is that the data used did not reveal the specific leadership practices that may have affected the school environment. Psychosocial safety climate (PSC) is a higher-level organisational construct that aligns with aspects of the systematic work environment management as it captures employees' collective perceptions of the extent to which organisational policies and procedures value, prioritise, and protect their psychological health and safety (Berthelsen, Westerlund et al., 2020; Hall et al., 2010). It is an organisational precursor to the JD-R model as it explains how the organisational climate provides the conditions for the psychosocial work environment factors and how it moderates the dual-pathway process in which psychosocial work environment factors impact employees' health and organisational behaviour (Zadow et al., 2019). Studies have shown that schools with a high PSC have organisational practices aimed at maintaining work demands at manageable levels, which results in better work-related health outcomes for teachers whereas in schools with low PSC teachers report higher levels of job demands and lower levels of recovery and psychological well-being (Dollard & Bakker, 2010; Garrick, Mak, et al., 2014; Garrick, Winwood, et al., 2014). These findings show that the influence of psychosocial work environment factors on teachers' work-related health is contingent upon the school's PSC (Garrick, Mak et al., 2014). However,

previous studies have not examined the association between PSC and teacher turnover. Such investigations are of practical importance for schools as they can support the notion of the systematic work environment management in schools being used as a tool to facilitate teacher retention (Bakker & Demerouti, 2017). Additionally, there is an over-reliance on teachers' self-reported turnover intention despite its weak link to actual turnover, and analysing its relationship with self-reported psychosocial work factors risks inflating associations due to common source bias (Bakker & Demerouti, 2017; Grant & Brantlinger, 2023; Podsakoff et al., 2003). As such, the lack of studies examining the relationship between schools' systematic work environment management practices and teacher turnover using objective turnover data represents a significant gap in the research literature (Bakker & Demerouti, 2017). This study aims to address the identified research gaps by examining the relationship of PSC and psychosocial work factors with teacher turnover rates at 12- and 24-months using objective turnover data from 14 schools in two Swedish municipalities. The contribution of our paper to the research literature is as follows: (i) we examine the relationship between school-level PSC and objective teacher turnover data; (ii) the use of school-level predictors such as PSC and group-level outcomes addresses important conceptual gaps in the literature and can inform the development of effective teacher retention interventions through the identification of stabilising organisational structures and practices; (iii) the merging of survey data with actual turnover data further strengthens the contribution of the study and guards against common method bias which is a prevalent weakness in the literature (Grant & Brantlinger, 2023).

2. Methods

2.1. Study design, population, and context

The present study uses data from a large randomized controlled implementation trial that examined the implementation of a guideline for the prevention of mental health problems among school personnel (Kwak et al., 2019). The trial was approved by the Swedish Ethical Review Authority (Dnr. 2017/984–31/5) and registered in ClinicalTrials.gov (NCT03322839, date assigned: August 2017). The data collection for the implementation trial occurred between September 2017 and September 2019 in 20 public primary schools. Of these, 8 schools were located in a metropolitan municipality with a large school market (~350 schools) whereas the other 12 schools were located in a rural municipality with a small school market (~20 schools). The schools in both municipalities provide a good representation of different geographical and socioeconomic catchment areas in Sweden. Additional ethical approval for the research questions in the present study was granted by the Swedish Ethical Review Authority (Dnr. 2022-04661-01). The present study used an open cohort repeated cross-sectional research design and merged teachers' self-reported survey data on psychosocial work environment factors and PSC at baseline ($n = 440$) and at a 12-month follow-up ($n = 406$) with annual turnover data for each school year. Six of the initial 20 schools were excluded from the present study as four were special education schools and the other two were undergoing extensive restructuring during the data collection period, in which one of them was to be subsumed by another school. As such, the data used in the present study are based on 14 schools, of which seven schools are in municipality 1 (urban) and the remaining seven schools are in municipality 2 (rural). The following sections will use the terms first school year, which models the association between baseline survey data and turnover at 12 months, and second school year, which models the association between survey data at 12 months and turnover at 24 months. School sizes, i.e., the number of teachers within each school unit, within the sample ranged from 15 to 57 during the first school year and from 16 to 55 during the second school year. As the study focuses on the school level, all teachers regardless of teacher certification status were included in the analysis. Table 1 provides a descriptive overview of the sample per school unit.

The Swedish public school system consists of voluntary preschooling and continues with nine years of compulsory education whereupon completion of the 9th grade provides the opportunity

Table 1. Descriptive overview of the sample per school.

	First school year ($n = 440$)					Second school year ($n = 406$)					Turnover rate	
School	*N*	School size[a]	Student composition (%)[b]	Teaching experience (%)[c]	Teacher certification (%)	*N*	School size[a]	Student composition (%)[b]	Teaching experience (%)[c]	Teacher certification (%)	12 months (%)	24 months (%)
1*	27	375	39	73.1	84.4	28	354	37	71.4	86.2	10	16.7
2*	21	281	16	90.5	94.7	19	325	18	77.8	82.6	16.2	12.8
3*	22	295	26	81.8	84.2	18	261	27	88.9	73.9	7.5	6.4
4*	52	810	21	68.6	83.9	55	508	23	81.8	93.3	20	29
5*	57	632	44	75.4	85.7	48	591	41	79.2	82.6	23.5	16.9
6*	31	420	45	80.6	78.1	33	417	43	72.7	71.7	18.9	14.3
7*	52	825	37	70.6	86.3	48	835	36	81.2	80.4	13.9	11.1
8	35	497	8	80	75.8	30	605	9	83.3	77.8	25.3	29
9	30	335	7	63.3	63.3	22	211	6	72.7	70	35.9	32.6
10	15	213	31	100	88.9	16	188	30	93.7	88.2	17.5	30.2
11	29	461	29	65.5	61.8	28	544	29	75	75.5	34.7	28.2
12	32	438	18	75	55.9	25	446	19	83.3	57.6	39.1	31.7
13	19	230	19	73.7	93.3	20	261	20	80	93.3	30	26
14	18	187	18	66.7	61.1	16	146	32	81.2	56.3	27.5	34.1

Note: *schools located in the urban municipality; [a] number of students at the school obtained from the Swedish National Agency for Education statistical database; [b] % of students with foreign background obtained from the Swedish National Agency for Education statistical database; [c] % of teachers in the sample with ≥5 years teaching experience obtained from survey data.

to advance to upper secondary schooling. Most public primary schools in Sweden are small and the average school consists of ~220 pupils, with only 10%–20% of pupils attending schools with 500 pupils or more (The Swedish Association of Local Authorities and Regions, 2018). The Swedish school market is characterised by school choice and competition between schools. The basic premise for this market adaptation is set by the state through educational policy and regulated in terms of the services provided by schools, the financial compensation tied to each pupil, and the terms for competition between schools (The Swedish National Agency for Education, 2012). The school market is also shaped by regional and municipal conditions as schools are funded by municipal grants and the municipality regulates the financial management and resource allocation of municipal schools. Although the municipality is responsible for the organisation and provision of school services, each school is responsible for their local school organisation (The Swedish National Agency for Education, 2013). Teachers' job opportunities are shaped by their local school market and wage dispersion is among the lowest in the Organisation for Economic Co-operation and Development (OECD) (OECD, 2019). Teachers are employed by the municipalities; however, there is no central municipal allocation of teachers and employment procedures are commonly handled by individual school principals. Teachers are employed on either a permanent or a fixed-term contract, with the former only available to certified teachers. Teachers are free to terminate their employment contract with a school subject to the contractual conditions which are either set during the employment procedure or specified by labour laws or union regulations. Reports by the Swedish School Inspectorate have shown that most municipalities lack a systematic approach to managing their teacher supply, which means that teacher retention and recruitment initiatives are often done on the school level (Swedish Schools Inspectorate, 2020).

2.2. Predictor variables

The Psychosocial Safety Climate questionnaire (PSC-12) was used to assess PSC using 12 items on a 5-point scale ranging from 1 = Strongly Disagree to 5 = Strongly agree (Hall et al., 2010). In line with the PSC-12 recommendations, the questions form a summative index out of 60 where a higher number indicates a better PSC at the workplace (Hall et al., 2010). The Cronbach alpha for PSC-12 was .94 in the first school year and .94 in the second school year, indicating good reliability (Nunnally, 1978). As the PSC-12 captures dimensions of the organisational climate and organisations' occupational safety and health practices it is often analysed as an aggregated measure of individual employees' perception of management's valuing and prioritising of their psychological well-being and safety (Berthelsen, Muhonen, et al., 2020; Zadow et al., 2019).

Psychosocial work factors were assessed using the Copenhagen Psychosocial Questionnaire II (COPSOQ II) (Pejtersen et al., 2010). The COPSOQ II is a validated and well-used instrument for psychosocial risk assessment at workplaces (Bjorner & Pejtersen, 2010; Leka & Jain, 2010). It can be well integrated into the JD-R framework (Berthelsen et al., 2018) and has previously been used to assess psychosocial risk factors in a school context (Dicke et al., 2018). The COPSOQ II was used to compute a job demands index consisting of three scales (*quantitative demands, emotional demands*, and *work pace*) and a job resource index consisting of five scales (*possibilities for development, social support from supervisor, social support from colleagues, recognition*, and *influence at work*), with each scale using a 5-point Likert scale ranging from 1 = Never to 5 = Always or from 1 = Very low to 5 = To a very high extent (Pejtersen et al., 2010). The Cronbach alpha for the job demand scales ranged from .77 to .84 in the first school year and from .77 to .83 in the second school year, indicating good reliability (Nunnally, 1978). The Cronbach alpha for the job resource scales ranged from .59 to .79 in the first school year and from .56 to .81 in the second school year, indicating acceptable to good reliability (Nunnally, 1978). Although the COPSOQ II collects responses at the individual level it is suitable for organisational risk assessment, as the scales can be aggregated to group mean scores to reflect collective perceptions of psychosocial work factors (Berthelsen, Muhonen et al., 2020).

2.3. Outcome variable

The outcome variable was objective annual teacher turnover rate measured at the school-level for both school years. Annual turnover is a short-term measure of turnover and operationalised as the loss of teachers (including unqualified teachers) from one school year to the next (Holme, Jabbar, et al., 2018). The annual teacher turnover rate for the schools was based on school employment records that detailed occupational role and the start and end of employment for all school personnel. For one of the participating municipalities the annual teacher turnover rate was calculated and provided by the municipality and for the second municipality it was calculated by the research team using school employment records provided by the municipality.

2.4. Control variable

The study also included two school-level control variables, teacher certification level and proportion of students with a foreign background, obtained from the Swedish National Agency for Education statistical database. We controlled for student composition (proportion of students with a foreign background) and teacher certification as studies in Sweden have shown that teacher turnover is higher in schools with higher proportions of students with a foreign background and low teacher certification levels (Holmlund et al., 2020). To rule out geographic and intervention effects, we also controlled for municipality type and intervention group. Municipality type was coded 1 (urban) and 2 (rural). Seven schools belonged to the intervention group, which was coded 1, and the remaining seven schools belonged to the control group, which was coded 2. The seven schools belonging to the control group were evenly matched according to municipality as three schools were based in the urban municipality and the remaining four schools were based in the rural municipality.

2.5. Statistical analyses

The study merges survey data collected at the individual level with objective turnover data, which is a group-level outcome. As such, the study represents a micro–macro data situation in which the dependent variable is measured at the group level and the independent variables are measured on the individual level (Foster-Johnson & Kromrey, 2018). Unlike macro–micro data situations, i.e., multilevel data situations in which the dependent variable is measured at the individual level, less work has been devoted to how to analyse multilevel data with group-level outcomes despite micro–macro data situations occurring in a variety of settings which are of interest to researchers within medical, social, and behavioural sciences (Croon & van Veldhoven, 2007). In general, this data situation can be approached using regression by either aggregation, in which data measured at the individual level are aggregated to the level at which the dependent variable is measured, or disaggregation, in which aggregated data are analysed at the individual level by repeating group variable scores for each individual (Bennink et al., 2013; Croon & van Veldhoven, 2007; Foster-Johnson & Kromrey, 2018). Both approaches come with statistical limitations in the subsequent regression analysis which are based on models that require variability in the data. Aggregating data to the group-level greatly reduces variability in the data and statistical power as the number of observations on which the subsequent regression analysis is carried out are decreased as groups are treated as the units of analysis (Bennink et al., 2013). Additionally, this approach leads to inaccurate estimates of standard errors of the regression parameters (Bennink et al., 2013; Croon & van Veldhoven, 2007; Foster-Johnson & Kromrey, 2018). Disaggregation allows for greater statistical power as the full sample can be utilised, as each individual in a group is assigned a group-level score for both the predictor and outcome variables which is subsequently treated as an independently observed score in a regression analysis. Although this enables greater statistical power it violates the assumption of independent errors among individuals and yields inaccurate estimates of the standard errors of model parameters (Bennink et al., 2013; Croon & van Veldhoven, 2007; Foster-

Johnson & Kromrey, 2018). Simulation studies of micro–macro data have shown that there are a variety of data conditions that must be considered when analysing micro–macro data prior to selecting the analytical approach (Foster-Johnson & Kromrey, 2018). The main consideration for the present study is the sample size in terms of the number of higher-level groups (i.e., schools) necessary to achieve statistical power and analytical precision to generate reasonable external validity of the results. In line with the pragmatism required when dealing with micro–macro data situations as suggested by Foster-Johnson and Kromrey (2018), and as the present study involves few higher-level groups ($n = 14$ schools), we decided to opt for the disaggregated approach in our analysis.

As the PSC-12 and COPSOQ II are instruments that are suitable for organisational risk assessment (Berthelsen, Muhonen et al., 2020), scale scores were aggregated from individual-level data to the school-level using group-mean centring, which refers to the subtraction of the group mean from each individual score to obtain a within-group average which provides a better group-level average as it corrects for confounding effects such as group size. The scale scores were then specified using an additive composition model which refers to the averaging of the lower-level variables (i.e., the sub-scales used to compute the Job Demands and Job Resource variables) to capture the higher-level variable (Chan, 1998). ICC(1) indices were calculated to justify aggregation (LeBreton & Senter, 2008). ICC(1) indicates the amount of variance in the teachers' responses that can be attributed to their school (Bliese, 1998; Bliese et al., 2019; Woehr et al., 2015). Typically ICC(2) indices are calculated when using aggregating variables as it provides an estimate of the reliability of group-level group means within a sample (Woehr et al., 2015); however. additive composition models do not require within-group agreement and therefore we opted to not calculate the ICC(2) (Chan, 1998; LeBreton & Senter, 2008). ICC(1) values within organisational research typically range up to .20, with values of .05 indicating small effects and higher values indicating a stronger effect. The ICC(1) for job demands was .01 in both school years and was .02 for job resources in the first school year and .04 in the second school year. The ICC(1) for PSC was .07 in the first school year and .11 in the second school year. Although the ICC(1) values for job demands and job resources did not support statistical aggregation, we decided to include the aggregated means for both indexes in line with the practical use of the COPSOQ for organisational risk assessment. Furthermore, the decision to use school-level predictors is in line with the recommendations made by Foster-Johnson and Kromrey (2018) that suggest that group-level analysis is the preferred strategy for micro–macro data situations. As such, the present study investigated the association of PSC, job demands, and job resources measured on the school-level with annual school-level turnover data for each school year. As we opted for the disaggregated approach, data were analysed using hierarchical regression.

3. Results

The overall means, standard deviations, and correlations in both school years are presented in Table 2. In the first school year, higher levels of job demands were correlated with higher levels of teacher turnover at 12months whereas higher levels of PSC, proportion of students with a foreign background, and teacher certification were correlated with lower levels of teacher turnover at 12-months. In the second school year, higher levels of job demands were correlated with higher levels of teacher turnover at 24-months whereas higher levels of job resources, PSC, proportion of students with a foreign background, and teacher certification were correlated with lower levels of teacher turnover at 24-months. Table 3 presents the results from the hierarchical regression analysis using the disaggregated approach. For both school years, the predictors were added in the first step followed by the control variables in the second step. PSC and psychosocial work factors, entered in step 1, explained 7% of the variance in teacher turnover at 12-months, $F(3, 436) = 10.55$, $p < .001$. The control variables entered in the second step explained an additional 79% of the variance in teacher turnover at 12-months, $F(7, 432) = 382.14$, $p < .001$. The total model explained 86% of the

Table 2. Descriptive statistics and school-level correlations between measured variables.

Variable	Mean	SD	Min–Max	1.	2.	3.	4.	5.	6.
First school year ($n = 440$)									
1. Job Demands	3.65	.12	3.36–3.89	1					
2. Job Resources	3.35	.11	3.18–3.71	−.39**	1				
3. Psychosocial Safety Climate	27.9	2.7	22–34	−.41**	.53**	1			
4. Student composition[a]	27.1	12.7	7–45	.07	−.06	.50**	1		
5. Teacher certification[a]	78.7	11.5	55.9–94.7	−.26**	.28**	.41**	.39**	1	
6. Teacher turnover 12-months[a]	22.8	9	7.5–39.1	.16**	.01	−.19*	−.47**	−.76**	1
Second school year ($n = 406$)									
1. Job Demands	3.58	.13	3.30–3.80	1					
2. Job Resources	3.38	.13	3.12–3.67	−.13*	1				
3. Psychosocial Safety Climate	29.6	3.4	24–38	−.19**	.78**	1			
4. Student composition[a]	27.9	11	6–43	−.35**	.15**	.18**	1		
5. Teacher certification[a]	79.4	10.3	56.3–93.3	−.57**	.24**	.28**	.06	1	
6. Teacher turnover 24-months[a]	22.1	8.5	6.4–34.1	.19**	−.19**	−.24**	−.61**	−.13**	1

Note: *correlation is significant at the 0.05 level (2-tailed); **correlation is significant at the 0.01 level (2-tailed). [a]Expressed as a percentage.

Table 3. Hierarchical regression results for psychosocial work factors and psychosocial safety climate on teacher turnover.

	First school year		Second school year	
Variables	Model 1	Model 2	Model 1	Model 2
Predictors				
PSC	−.23** (.19)	.70** (.12)	−.20* (.20)	−.46** (.13)
Job Demands	.13* (3.80)	.09** (1.77)	.15** (3.30)	−.08* (2.21)
Job Resources	.18** (4.74)	−.90** (4.31)	−.01 (4.97)	−.12* (2.66)
Controls				
Municipality		1.12** (.91)		.82** (.70)
Intervention group		.47** (.79)		.32** (.78)
Teacher certification		.23** (.04)		.40** (.03)
Student composition		−.48** (.03)		−.23** (.03)
R^2	.07	.86	.08	.76
Adjusted R^2	.06	.86	.07	.75
F	10.55	382.14	11.07	178.58
$R^2\Delta$	.07	.79	.08	.68

Note: standard errors are within parentheses. *Significant at the 0.05 level (2-tailed); **significant at the 0.01 level (2-tailed).

variance in teacher turnover at 12-months: $F(7, 432) = 382.14$, $p < .001$, $R^2 = .86$, $R^2_{\text{Adjusted}} = .86$. The strongest predictor of turnover at 12-months was municipality ($\beta = 1.12$), followed by job resources ($\beta = -.90$) and PSC ($\beta = .70$). In the second school year, PSC and psychosocial work factors, entered in step 1, explained 8% of the variance in teacher turnover at 24-months, $F(3, 402) = 11.07$, $p < .001$. The control variables entered in step 2 explained an additional 68% of the variance in teacher turnover at 24-months: $F(7, 398) = 178.58$, $p < .001$. The total model explained 76% of the variance in teacher turnover: $F(7, 398) = 178.58$, $p < .001$, $R^2 = .76$, $R^2_{\text{Adjusted}} = .75$. The strongest predictor of turnover at 24-months was municipality ($\beta = .82$), followed by PSC ($\beta = -.46$) and teacher certification ($\beta = .40$).

4. Discussion

The current study examined the relationship of PSC and psychosocial work factors with teacher turnover rates at 12- and 24-months using objective turnover data from 14 schools in two Swedish municipalities. The results for both school years show that PSC and psychosocial work factors explain 7% and 8% of the variance in annual teacher turnover. In the null model for both school years, higher levels of PSC were associated with lower levels of annual turnover. In the first school year, the association between PSC and annual turnover changed in the second step of the

regression analysis when the control variables were added, and surprisingly higher levels of PSC were associated with increased annual turnover. This unexpected finding can potentially be attributed to the inadequate implementation of systematic work environment management practices (Frick, 2016). Although PSC captures employees' collective perceptions of the extent to which organisational policies and procedures value, prioritise, and protect their psychological health and safety it is important to distinguish between espoused PSC, i.e., what organisations claim to do in their policies and practices, and enacted PSC, i.e., what organisations actually do (Dollard et al., 2010; Yulita, Dollard, & Idris, 2017). For instance, the reports by the Swedish Work Environment Authority highlighting the inadequate implementation of the mandatory systematic work environment management practices in schools and school-based work environment intervention studies provide evidence for the discrepancy between schools' espoused and enacted systematic work environment practices (The Swedish Work Environment Authority, 2017; Toropova et al., 2022, 2023). In the second school year, the observed association between PSC and lower levels of annual turnover was maintained despite the addition of the control variables to the model. The observed strong relationship in which higher levels of PSC were associated with lower levels of annual turnover provides support for the notion of the mechanisms underlying teacher retention aligning with the objectives of psychosocial risk management (Madigan & Kim, 2021). As PSC is a precursor to psychosocial work factors it can facilitate teacher retention by reducing psychosocial risk factors in the work environment, which is central to reducing teacher turnover (Madigan & Kim, 2021). In line with findings in the teacher retention literature (e.g. Björk et al., 2019; McCarthy et al., 2009), our results also showed that higher levels of job resources were associated with lower levels of annual turnover in both school years. This effect was observed in both school years but was more strongly associated with turnover in the first school year. Given that higher levels of PSC were associated with higher levels of annual turnover in the first school year, potentially due to the discrepancy between espoused and enacted PSC, it is interesting to note that job resources were particularly important for better teacher retention during this period. It is possible that job resources were particularly important for teacher retention during this period due to the lack of top-down processes influencing the organisation of teachers' work. This notion is somewhat supported by the findings in the second school year as the influence of job resources on teacher retention lessens and is seemingly replaced by that of PSC. This is in line with, and provides support for, research in the teacher retention literature that has shown that teachers use a range of proactive strategies that enable them to amass protective resources that can in turn facilitate teacher retention by safeguarding teachers from the health-impairing influence of psychosocial risk factors in the work environment (Brunetti, 2006; Castro et al., 2010; Pomaki et al., 2010). In line with findings in the teacher turnover literature, higher levels of job demands were related to higher levels of annual turnover during the first school year; however, an increase in job demands was associated with lower annual turnover in the second school year, although the effect observed was small.

In the present study, annual teacher turnover rate in both school years was mainly explained by municipality type, and to a smaller extent by PSC and psychosocial work factors. Much of the teacher turnover literature is focused on schools in urban environments and disregards the teacher retention challenges that many schools in rural communities face (Ingersoll & Tran, 2023). The teacher retention challenges facing rural schools can be viewed as a form of spatial injustice, as the teacher staffing difficulties in these schools can be explained by labour market dynamics that are embedded and maintained through geography (Tieken, 2017). The data in the study, albeit a small sample, illustrate the regional teacher supply differences which have been raised by The Swedish National Agency for Education (2023) as the average teacher certification level in the rural municipality was ~69% compared to ~85% in the urban municipality. Therefore, a potential reason for the strong influence of municipality type on turnover could be due to schools situated in rural areas relying on a higher proportion of unqualified teachers who are employed on fixed-term contracts and thus more prone to turnover (Cheng & Chan, 2008). Our

findings showing that the teacher turnover rate was higher in schools in the rural municipality are in line with international studies showing that many rural schools suffer from a revolving door of teachers (Harbatkin, 2023; Ingersoll & Tran, 2023). Furthermore, these findings contrast with previous studies conducted in Sweden which have suggested that teacher turnover is higher in schools situated in urban communities, as teachers in these schools have access to a larger school market which can facilitate mobility (Holmlund et al., 2020). In addition to municipal effects, the findings also showed an effect of the intervention grouping as schools in the control group had higher levels of turnover. Although the seven schools belonging to the control group were evenly matched according to municipality, it is possible that this effect was driven by the four rural schools in the control group. Teacher certification was also a strong predictor of teacher turnover which further strengthens the notion of labour market dynamics playing a key role in schools' staffing difficulties. Moreover, as our turnover measure included unqualified teachers, it is possible that the increased turnover reflects the mobility of the unqualified teachers. It could be argued that the inclusion of unqualified teachers inflates the turnover rate observed in the schools; however, this provides a more comprehensive image of the teacher staffing difficulties that schools face (Donitsa-Schmidt & Zuzovsky, 2016). Furthermore, a contributing factor to the deleterious effects of teacher turnover on the functioning of schools is the loss of school-specific human capital (Gibbons et al., 2021), which is also amassed by unqualified teachers. Similarly to previous studies conducted in Sweden, our results also show that schools with a higher proportion of students with a foreign background do not seem to experience higher turnover rates (Karbownik, 2014, 2020). This finding is in stark contrast to the results from international studies (e.g. Barbieri et al., 2011; Simon & Johnson, 2015), and can be explained by the compensatory resource allocation mechanism embedded within the Swedish school system wherein additional resources can be allocated by municipalities to schools based on their local conditions and needs (The Swedish National Agency for Education, 2013). Studies have shown that the compensatory resource allocation mechanism has led to a general increase in teacher density in schools with a high proportion of students with a foreign background (Hansson & Gustafsson, 2017), although it is important to note that the teachers in these schools tend to be less experienced (Swedish Schools Inspectorate, 2020; The Swedish National Agency for Education, 2023). Furthermore, it is also important to note that the compensatory resource allocation mechanism also involves increasing the number of paraprofessional support roles (e.g., teacher assistants and student assistants) and these support roles are sometimes used as unqualified teachers (The Swedish National Agency for Education, 2013). Based on our findings, differences in teacher turnover between schools in our study are mainly explained by municipal effects rather than differences in the management of psychosocial risk factors. Our results do however suggest that PSC is a viable predictor of teacher turnover despite its influence being overshadowed by labour market dynamics. Lastly, it is important to note that turnover is a natural part of the organisational life cycle and is a mechanism that can refresh the human capital in schools as well as facilitate better person–organisation matches among teachers. However, high levels of turnover are undesirable as this has lasting effects on the instructional quality of the teaching staff and student achievement (Sorensen & Ladd, 2020). Furthermore, it can trigger a cycle of negative effects in organisations as it can catalyse additional teacher turnover, prompting further instability for schools (Holme, Jabbar, et al., 2018). Although improving teacher retention by fostering healthy workplaces should be a key objective for schools, our results implicating labour market dynamics also highlight how staff instability in schools intersects with teacher availability. Although the present study did not include longitudinal measures of turnover, which would offer a better understanding of school instability (Holme, Jabbar, et al., 2018), our findings still highlight the need for targeted recruitment and retention strategies in rural areas to prevent chronic instability in rural schools and further geographical disparities.

4.1. Methodological considerations

We acknowledge the need to interpret the study's findings with caution given the study's methodological limitations. Although the PSC and COPSOQ instruments are suitable for organisational risk assessment and can be analysed as an aggregated measure to reflect collective perceptions, aggregation greatly reduces variability in the data. Furthermore, aggregating data to the school-level disregards the heterogeneity within schools and is susceptible to measurement and sampling error (Bennink et al., 2013). The low ICC(1) values further raise questions about the validity and reliability of aggregating individual-level data. However, the scales used to compute the job demands, job resource, and PSC indexes all demonstrated good reliability prior to aggregation. In addition to this, the present study used survey data from a large randomised controlled implementation trial that was conducted on all school personnel in the selected schools with a total participation rate of 83.1%, which reduces the potential issue of sampling errors (Toropova et al., 2022). The lack of studies conducted on the school-level represents a significant gap in the research literature and aggregating data to the school level, despite its statistical limitations, is needed to further the understanding of between-school differences in teacher turnover. Furthermore, although merging survey data with objective turnover data prevents the issue of common method bias and is a considerable strength of the study, this approach comes with statistical challenges. There are few methods for analysing micro–macro data situations and the common approaches, disaggregation and aggregation, are not without significant statistical limitations. The present study used the disaggregation approach, due to insufficient higher-level groups (n = 14 schools), which violates the assumption of independent errors among individuals, yielding inefficient estimates and biased estimates of the standard errors of model parameters (Bennink et al., 2013; Croon & van Veldhoven, 2007). Moreover, this approach does not correct for the dependency among the individual observations within a group, which artificially inflates the statistical power (Bennink et al., 2013; Croon & van Veldhoven, 2007). It would have been preferable for the present study to have used the aggregation approach, despite its limitations, as simulation studies have shown that it performs well in comparison to more advanced approaches that have been introduced in recent years (Foster-Johnson & Kromrey, 2018).

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Data availability statement

The dataset used in the current study is not publicly available in line with the Swedish ethical review approval.

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